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Reactions to the Bernanke Review from Bank of England watchers☆

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ABSTRACT

We summarize reactions to the Bernanke Review from the Bank of England watchers community – a diverse group of academics, market economists, and business analysts who closely monitor and analyze the actions of the Bank of England. Key themes include the Review's recommendations to retire the "fan chart", increase the use of scenario analysis, and de-emphasize the central forecast conditioned on the market yield curve, as well as its critique of the Bank's forecasting infrastructure. There is also extensive discussion of areas left unaddressed by the Review, including whether the Monetary Policy Committee should publish its preferred policy rate path, adopt a Fed-style dot plot, or give Bank staff ownership of the forecast.

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1. Introduction

On 12 April 2024, Ben Bernanke, Nobel Laureate and former Chair of the Federal Reserve Board, published his landmark review into "Forecasting for Monetary Policy Making and Communication at the Bank of England" (Bernanke, 2024). Bernanke's remit for this review was to consider "the appropriate approach to forecasting and analysis in support of decision-making and communications in times of high uncertainty from big shocks and structural change". Bernanke's review drew on a broad evidence base including the academic literature on macroeconomic forecasting and interviews with key internal and external stakeholders; he also attended key meetings at which the Monetary Policy Committee (MPC) discussed and finalized its forecast and policy decisions, and he compared the forecasting procedures and records of the Bank of England (henceforth "the Bank") with those of six peer central banks. The Bank welcomed the review and committed to action on all twelve of Bernanke's recommendations.

We solicited reactions to the Bernanke Review from a set of prominent UK macroeconomists, which we subsequently published in an e-book that can be accessed via the King's College London website (Aikman & Barwell, 2024). This included both academics and economists working in financial markets and business; many were former members of the MPC or former Bank staff economists with experience in the forecast and model development process. Each author had complete discretion over which aspect of the Review to focus on.

In this paper, we summarize these reactions to the Bernanke Review from the Bank watchers community. They fall into five common themes: the demise of the "fan chart"; the increased use of scenario analysis; conditioning assumptions in the central forecast; MPC versus staff ownership of the forecast; and the failure of the Bank's forecast infrastructure, including its main macroeconomic forecasting model, "COMPASS". We present the

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arguments in an unfiltered way, as the authors themselves have expressed them – in this sense, any errors are those of the contributors rather than our own. But in writing this piece, we have of course been selective in choosing which arguments to emphasize, an editing process for which we take full responsibility.

The paper is organized as follows. Section 2 provides a brief reprise of the twelve recommendations of the Bernanke Review. Section 3 presents reactions from the watcher community to these recommendations and to the review more generally. Section 4 concludes.

2. The Bernanke Review's recommendations

The Bernanke Review makes twelve recommendations covering (a) the infrastructure the Bank uses for forecasting and analysis; (b) the use of the forecast in the MPC's decision-making process; and (c) the communication of the MPC's outlook and policy rationale to the public.

The recommendations are as follows[1]:

R1. The ongoing updating and modernization of software to manage and manipulate data should be continued with high priority and as rapidly as feasible.

R2. Model maintenance and development should be an ongoing priority, supported by a significant increase in dedicated staff time and adequate resources, including specialized software as needed.

R3. Over the longer term, the Bank should undertake a thorough review and updating of its forecasting framework, including replacing or, at a minimum, thoroughly revamping COMPASS (the Bank's existing macroeconomic model).

R4. Based on the lessons of recent years, a revamped forecasting framework should include at least the following key elements: (a) rich and institutionally realistic representations of the monetary transmission mechanism, allowing for alternative channels of transmission; (b) empirically based modeling of inflation expectations, with a distinction between short-term (e.g., one year) and longer-term (e.g., five to ten years) expectations, and without the assumption that longer-term inflation expectations are always well-anchored; (c) models of wage–price determination that allow gradual adjustment and causation from prices to wages as well as from wages to prices; (d) detailed models of the financial sector, the housing sector, the energy sector, and other key components of the UK economy; (e) greater attention to, and ongoing review of, supply-side elements and their role in the determination of inflation and growth. Important supply-side factors include changes in productivity, labor supply, the efficiency of job–worker matching, supply-chain disruptions, and trade policy. Notably, analyses of inflation should consider supply-side factors as well as the state of aggregate demand.

R5. Incrementalism (the practice of basing new forecasts on previous forecasts, with marginal adjustments) and the use of ad hoc judgments may obscure deeper problems with the underlying forecasting framework or unrecognized changes in the structure of the economy. The staff should be charged with highlighting significant forecast errors and their sources, particularly errors that are not due to unanticipated shocks to the standard conditioning variables. Models and model components that may have contributed to forecast misses should be evaluated and discussed regularly, as well as the determinants of variables whose forecasts are consistently dominated by extra-model judgments. Staff should routinely meet with MPC members to consider whether structural change, misspecification of models, or faulty judgments warrant discrete changes to the key assumptions or modeling approaches used in forecasting.

R6. The Bank should review its personnel policies to determine if existing staff could be deployed in ways that improve the forecasting infrastructure and forecast quality.

R7. To improve the MPC's policy discussion, the central forecast should be augmented regularly by alternative scenarios, with the specific scenarios ideally decided upon at an early stage of each forecast round by the MPC and staff.

R8. The publication of selected alternative scenarios in the Monetary Policy Report (MPR), along with the central forecast, would help the public better understand the reasons for policy choices, including risk management considerations.

R9. Because the standard conditioning assumptions do not necessarily reflect the MPC's views but can have potentially significant effects on the forecast, and because the central forecast by itself does not provide a clear rationale for policy decisions, the MPC should deemphasize the central forecast based on the market rate path in its communications and be exceptionally clear in warning about situations in which it judges the standard conditioning assumptions to be inconsistent with its view of the outlook.

R10. To put less emphasis on the central forecast, to simplify its policy statement, and to reduce repetitiveness in its communications, the MPC should replace or cut back the detailed quantitative discussion of economic conditions in the Monetary Policy Summary in favor of a shorter and more qualitative description, following the practice of most peer central banks.

R11. Despite their distinguished history, the fan charts as published in the MPR have weak conceptual foundations, convey little useful information over and above what could be communicated in other, more direct ways, and receive little attention from the public. They should be eliminated. Mean forecasts as currently constructed do not provide additional useful information and should also be dropped from publications in favor of more qualitative descriptions of risks and uncertainty surrounding the outlook.

R12. A phased approach to implementing changes proposed in this report, focused first on improving the forecasting infrastructure, while moving cautiously in adopting changes to policymaking and communications, is likely to be necessary. To facilitate infrastructure improvements and address existing deficits, the commitment of additional resources will be required, at least for a time.

[1] For conciseness, we repeat only the bold text from the Bernanke Review's recommendations – except for Recommendation 4, which we repeat in full.

3. Reactions from the BoE watchers community

These recommendations elicited a variety of opinions from the Bank of England watchers who contributed to our compendium. In what follows, we summarize what we see as the key themes running through these contributions, focusing on aspects relevant to forecasting.

3.1. The demise of the fan chart

A central theme in the Bernanke Review is the challenge of calculating and communicating the uncertainty underlying economic forecasts. In her contribution to our book, Kate Barker reminds us that this uncertainty includes whether the known shocks that have just hit the economy will play out as expected; the outcome of known future events – such as an election; and future unknown shocks (or unknown unknowns, as Donald Rumsfeld once put it).

The Bank's current practice of communicating uncertainty centers around the so-called "fan chart" diagram, a graphical representation used to illustrate uncertainty in economic forecasts (Britton, Fisher, & Whitley, 1998). This has been a central feature of the Bank of England's Inflation Report (since re-named the Monetary Policy Report) since the late 1990s. It provides a probabilistic depiction of the forecast ranges, helping to communicate the inherent uncertainty in projections.

Fig. 1 presents an example of this diagram, taken from the November 2019 Monetary Policy Report. The colored bands in the chart represents areas with the same probability of occurring; the dark band contains the modal forecast, with outcomes in the lighter shaded bands having lower probability.[2] While the width of the fan charts is informed by the experience of past forecast errors, their shape ultimately reflects the MPC's (collective) subjective assessment of the second and third moments of the distribution of outcomes.

The Bernanke Review offers a severe crtique of the continued use of fan charts, arguing that they "suffer from significant analytical weaknesses and have outlived their usefulness". The Review's main concern is that the width and skew of the fan chart appear to have "little or no explicit grounding in data or quantitative analysis and thus convey a false sense of precision" and are therefore "uncomfortably ad hoc". Thus, the review argues that "it is highly questionable whether fan charts provide any information that is not already conveyed (with less intervening noise) by the historical distribution of forecast errors and by qualitative MPC statements". As the mean forecasts inherit the conceptual weaknesses of the fan charts, they too should be retired as they "add little to the MPC's qualitative characterizations of risk and uncertainty". Finally, while retaining the fan charts might be justified "if they served a useful pedagogical purpose about the high uncertainty associated with forecasts", in practice the charts "receive little attention from the public and are rarely shown in media reports".

There were mixed reactions from the watchers we surveyed to the recommendation to retire the fan charts. Most agreed that the fan chart technology, in its current form, was in need of a refresh. One especially interesting critique came from Kate Barker, who noted that the width of the fan charts only seemed to increase over time. Barker explained, "While I was on the committee, it never seemed appropriate to say we were less uncertain". This observation highlights a potential limitation of basing the shape of the fan chart on MPC members' subjective assessments: that there might be a bias toward reflecting increasing uncertainty even when circumstances might reasonably allow for a more confident forecast.

This notwithstanding, most commentators who wrote on this matter felt that the problem was in the estimation rather than the presentation, and that the fan charts were worth retaining. Jagjit Chadha argues that "if used properly, [fan charts] can convey the balances and magnitude of risks that face policy makers and can actually act to complement scenario analyses". Many noted that the fan charts are particularly well suited for illustrating that the central forecast will inevitably be wrong, or as Ben Nelson put it, "stuff happens". Others saw value in the regular process of the MPC discussing and agreeing on these charts: "they do at least force the MPC to come to a common view around economic uncertainty, and the skew of risks", Ben Nabarro writes. Richard Barwell notes that the recommendation to abandon a mean projection (alongside eliminating the fan) in favor of a mode and a handful of inevitably arbitrary scenarios amounts to "a retreat from orthodoxy on optimal policy", which may be especially problematic in "atypical circumstances when a central bank switches into risk management mode".

On Bernanke's critique that the fan charts receive little public attention, Charles Goodhart argues that this is "as much a failing of the public and the media as it is of the approach". The fundamental problem he suggests is that the Review "does not appreciate that the public, politicians and media are looking to the forecasts to give them more guidance and certainty about what will happen, i.e. the central forecast, and are less concerned about possible risks and uncertainty". For this reason, "the scenarios would just be disregarded by public and media, as was the fan, concentrating on the middle, central projection". This remark highlights the challenge in communicating uncertainty, whether through fan charts, scenarios, or other means.

3.2. Scenario analysis

As a replacement to the fan chart, the Bernanke Review recommends publishing alternative scenarios alongside the central (modal) forecast.

These scenarios, the review explains, might include those that "(1) allow for direct comparisons of the likely effects of alternative policy paths on the outlook; (2) help to assess the effects and costs of possible risks to the outlook arising from unexpected changes in exogenous variables; (3) can be used to evaluate the effects of the Committee's policy choices on the economy if one or more of its key assumptions about the structure of the economy

[2] See the discussion in the box on pages 48–49 of the May 2002 Inflation Report for further details (Bank of England, 2002).

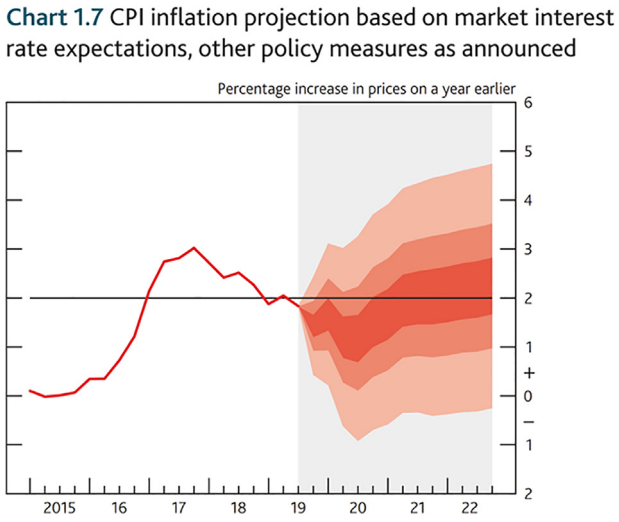


Fig. 1. Fan chart for CPI inflation from the November 2019 Monetary Policy Report.
Source: Bank of England (2019).

are wrong; and (4) can be used to decompose historical forecast errors into portions due to judgements, conditioning assumptions, and other factors". The review goes on to note that "selected alternative scenarios could also provide the public with information about the Committee's policy reaction function and its view of the monetary transmission mechanism".

There was qualified support for this recommendation among Bank watchers. Peter Westaway notes that "it is difficult to argue with the suggestion that forecasts offering alternative views of how the economy might evolve could help elucidate the policy outlook". Mick Grady argues that "publishing a number of alternative scenarios can indeed be helpful in articulating the uncertainty and the balance of risks … It would add to the richness of the policy debate and should help with communications". Tomasz Wieladek contends that scenario analysis can enhance credibility: "By presenting alternative scenarios before they happen, the central bank can demonstrate that it understands the root causes of inflation in a particular state of the world. This will instil the public with confidence that the central bank will return inflation to target and support the anchoring of long-term inflation expectations at target".

However, many watchers emphasized the lack of a solid methodology for generating scenarios and questioned whether the Bank's modeling toolkit is up to the task. David Hendry and John Muellbauer caution that "A scenario based on a bad model that would shift when the scenario change is implemented is often worse than useless"; moreover, they cite the result in Hendry and Pretis (2023) that scenarios are prone to serious problems. David Aikman highlights the potentially significant practical problems that the Bank may face in producing alternative scenarios, particularly given the issues that Bernanke has raised regarding the current state of the forecasting toolkit at the Bank: "can the Bank's models be relied upon to provide an accurate gauge of the behaviour of macroeconomic variables under the assumed scenario? Recall that the Bank doesn't use its macroeconomic model at all at present to generate the existing fan charts … suggesting it has some doubts about the reliability of its toolkit for conducting such exercises".

Some of our contributors remind us that the use of scenario analysis in monetary policy analysis is hardly new. Adrian Pagan observes that "central banks have always done such scenarios in order to gauge the risks around a forecast. I do recall in the early 2000s that the process in place for deciding on the final projection asked the MPC what they saw as the three big risks to it and the staff then needed to investigate the sensitivity of the projection to them". And Sanjay Raja notes that "Dr Stockton proposed this more than a decade ago when conducting a review of the BoE's forecasting capabilities". Others, however, see the potential for a more profound change in practice. James Smith argues that "*if* producing scenarios means putting alternative projections at the heart of a revamped forecast round, with meaningful discussion of the accompanying policy response, then this would indeed be a big change". "Would this have changed anything during the cost-of-living crisis period?" he asks. "Arguably, yes. In 2022, for example, the MPC could have signalled that much higher inflation would have been accompanied by much higher rates. Doing so would have forced MPC to discuss that scenario in depth as part of its forecasting round, and to have communicated the implications explicitly to markets and the public".

Two further key themes emerge from the Bank watchers' responses on this topic.

The first concerns the choice of scenarios, the potential number of which, Charles Goodhart warns, is "huge, and the choice of which scenario to adopt is, surely, even more ad hoc than the fan chart". Ben Nelson observes that one option for doing so is to leverage the list of key judgments the MPC already publishes in its Monetary Policy Summary. This would amount to "quantify[ing] the economic significance of these judgements, putting

quantitative markers down for how the Committee intends to assess whether the judgements it has made are tracking or not". This would have the additional benefit of "provid[ing] an ex ante framework within which quantitatively to interpret forecast errors ex post". Ben Nabarro highlights the risk that shifting to a process organized around multiple scenarios could undermine the process of reaching "a common view around economic uncertainty, and the skew of risks … taking us further away from a common strategic discussion". While this is an important concern, it is also possible that considering multiple scenarios could enhance rather than detract from strategic discussions by broadening the range of uncertainties considered.

A deeper concern highlighted by the watchers we surveyed is that we lack a solid methodology for evaluating the veracity of chosen scenarios ex post. For instance, David Aikman asks, "How should the MPC be held to account ex post for its choice of scenarios? With fan charts, one could at least ask whether over long periods of time inflation outturns are consistent with the various probability bands of the forecast. It's not at all clear how this can be done with scenarios". These problems might be amplified by strong institutional and cultural influences at the Bank, which may blind policymakers to diverging opinions that might highlight unforeseen risks. While Bernanke discusses this problem of "group think" extensively in his review, DeAnne Julius expresses the concern that the recommendations "might have the unintended consequence of reinforcing 'group think' by further constraining the time and ability of external MPC members to bring outside perspectives into the process".

A complication emphasized by several commentators is that many salient risk scenarios are inherently difficult to communicate publicly. Geopolitical risks are an example here. Charles Goodhart notes that "an obvious risk to world economic conditions is…the outcome of the future US Presidential Election". He then asks: "even if the Bank of England was to consider this internally, could it give two risk scenarios with a Biden or a Trump victory?" Similar issues apply to the assumed path of fiscal policy in the forecast. Kate Barker writes that "adopting different paths for fiscal policy, though occasionally desirable, would present very evident difficulties (is the Bank recommending this path? Does the Bank have inside information?)".

The second theme relates to the assumed behavior of monetary policy in the chosen scenarios. Several commentators make the case for the MPC explaining how it would likely respond were a scenario to materialize. For example, Ben Nelson argues that "because monetary policy attains its stabilization power via its systematic responses to inflation prospects, using scenarios to promote a greater understanding of the Committee's reaction function in this way could enhance the effectiveness of policy". He argues that the mechanism by which this would operate would be "encouraging a more efficient response of market pricing to unexpected developments". Richard Barwell agrees, arguing that "the value of these scenarios will be limited if we don't know what is happening to monetary policy in each scenario. The differences in the paths for output and inflation are very hard to interpret if you don't know the underlying differences in interest rates and asset prices (e.g., the exchange rate)".

However, other MPC watchers highlight the challenges that come with implementing this approach. DeAnne Julius warns of the difficulties of the nine MPC members in agreeing what that preferred response should be: "[the MPC's policy response to a scenario] would represent a strong form of forward guidance that the Committee would find difficult to agree even if it were minded to provide it (which I doubt). It is no accident that monetary policy decisions become more data-dependent, and less forecast-dependent, when major shocks occur". Charles Goodhart highlights the problem that allowing for an endogenous monetary policy response might significantly over-state the MPC's stabilization powers. He asks, "when developments in conditioning variables go astray, how soon should one assume that the central bank reacts? How far should one assume that the Central Bank has correct knowledge of the new, revised future path of that conditional assumption? Assuming that the Central Bank had an exact appreciation of the likely revised path of the conditioning assumption would imply far less deviations than would be likely in practice".

3.3. Conditioning paths for interest rates

Turning to the MPC's central (i.e., modal) forecast, its current practice is to publish two projections, each of which makes a different assumption about the path of monetary policy over the twelve quarters of the forecast. The first is conditioned on the future path of short-term interest rates implied by money markets; the second assumes that short-term interest rates remain constant. The former projection is typically given prominence in the Monetary Policy Summary.

The Bernanke Review argues that there are "significant disadvantages" with this practice. Most importantly, "if the MPC in fact has a different view than the market about the likely course of policy, then a forecast conditioned on the market curve may be misleading". Moreover, "if the MPC policy action and communication cause the market curve to shift, as it likely will if the policy action deviates from market expectations, then the market-based conditioning assumption becomes immediately out of date (Goodhart, 2009)".[3] Despite these disadvantages, the Review falls short of suggesting a wholesale change in approach. Bernanke argues that change "would be highly consequential" and "recommends leaving decisions on this issue to future deliberations". Instead, he recommends that, in circumstances where the market path materially differs from the MPC's expectations, the Committee should be clear about this, and greater focus should be placed on scenarios that de-emphasize the market curve projection.

[3] The Bernanke Review also highlights several more technical difficulties associated with the practice of conditioning forecasts on the market curve. These include the challenge of stripping out risk and liquidity premia from market prices, the potential for the use of four-week-long windows of market price data to result in the inclusion of stale quotes, and the fact that market prices will presumably predict mean rather than modal values for Bank Rate.

These recommendations drew a variety of opinions from the Bank watchers community. For many, this was a missed opportunity to push the MPC to follow the practice of other leading central banks and begin communicating its expected outlook for Bank Rate. For example, Jens Larsen observes "it is not clear whether the reluctance to push for change is driven by the infrastructure inadequacies or by MPC resistance to embrace such a change. Either way, it is an opportunity lost: the argument in favour of this change has long been settled, and the review should have made such a recommendation". Chris Giles labeled the entire review "worse than a missed opportunity" and argues that by leaving the "highly consequential" decision to future deliberations, Bernanke's review is itself "inconsequential".

The Bank could communicate about the future path of policy either by publishing its collectively agreed preferred policy rate path over the forecast as per the practice of the Sveriges Riksbank and the Norges Bank.[4] Or it could publish something akin to the "dot plot" employed by the Federal Reserve to show individual FOMC members' views of the appropriate rate path. Bernanke addresses the case for the latter in a footnote of his review and concludes that "this approach would have the substantial drawback of uprooting the longstanding collaboration of staff and MPC members in the development of a common Bank forecast, and for that and other reasons is not recommended". Fig. 2 presents examples of both approaches.

The Bank watchers community highlighted several key arguments in favor of the MPC publishing its preferred policy path. Richard Barwell argues that this would reveal "significant unpublished information – about the Committee's assessment of the economy and the appropriate reaction function – and therefore its publication has the potential to move market expectations". He argues too that publishing a path is "a more disciplined and effective way to move those expectations (and anchor financial conditions in the 'right' place) than the current approach of hints and qualitative signals". Last, he argues that a commitment to publish a path "would focus the internal discussion on what really matters (monetary strategy) improving the quality of the internal debate". Mick Grady also emphasizes this latter point, arguing that such a move would "improve the internal policy debate (including on monetary strategy, which goes beyond the immediate policy decision) and improve communications by providing a clearer description of how the immediate and likely future policy decisions would bring inflation back to target". For Jagjit Chadha, the key benefit is that the publication of interest rate paths or fed-style dot plots "would provide much more clarity on the management of risks". And Jumana Saleheen tells us that "markets love the dot plots", which are "easy to communicate, easy to read the forecast numbers (unlike the fan charts), easy to see the range of views, and changes to the dots over time". She also highlights the fact that MPC members revealing their individual short and longer-run forecasts for inflation, growth and policy rates "is part of the MPC's individual accountability to both the UK Parliament and the Public".

To be clear, however, there is not universal support for this idea amongst the watchers we surveyed. One oft-heard criticism, here expressed by Adrian Pagan, is that the MPC's preferred path will come be treated as a promise: "As we have seen in Australia, it is easily the case that, when the monetary authority suggests a likely interest rate path, it starts being treated as what will happen, and decisions are made on that". Peter Westaway highlights a more technical reason for favoring conditioning on the market curve, which is that "it is this very path which economic agents will be basing their current and future decisions on, and which will be embodied in asset prices like the exchange rate, equity prices and bond yields". For this reason, he sees the current approach as "an informative way, perhaps the best way, to frame the policy decision". However, DeAnne Julius cautions that "few in the general public have any understanding or exposure to the market's yield curve, so using it as a conditioning assumption merely confuses most of the lay audience, including many journalists". She suggests placing greater emphasis on the constant rate forecast as this "is more intuitive for many people – and easier to explain – because it shows in which direction rates are likely to move (all else equal)".

3.4. MPC versus staff forecast

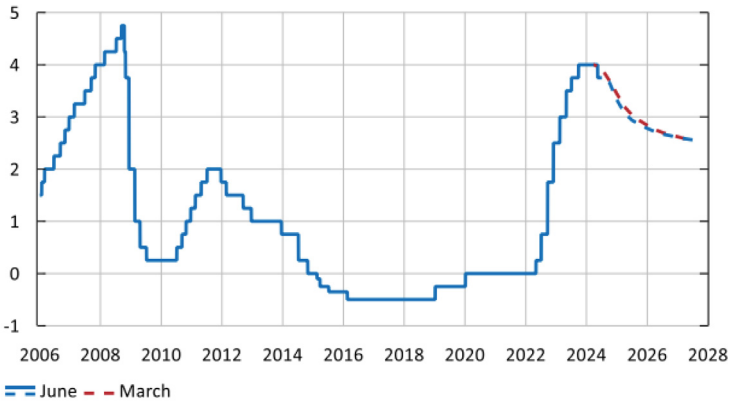
The Bernanke Review notes that the Bank of England differs from the majority of its peers in that the MPC plays an extensive role in the construction of the forecast, with the published forecast representing the best collective judgment of the Committee, although interestingly Bernanke speaks of "the longstanding collaboration of staff and MPC members in the development of a common Bank forecast". This forecast also plays a highly prominent role in the Committee's policy deliberations and communications. In this sense, the MPC's practice perhaps comes closest to the Platonic ideal of the "inflation forecast targeting" approach long advocated by the academic literature (see, e.g., Svensson, 1997). In contrast, the macroeconomic forecasts of the ECB and the Federal Reserve Board are produced by the staff, with little or no policymaker input (although FOMC members publish their own forecasts alongside every second policy decision); while at the Bank of Canada, the Riksbank, and the Norges Bank, policymakers sign off on the forecast but their involvement in its construction is rather limited.

Helen Thomas puts her finger on a fundamental tension with the forecast seemingly serving two distinct roles that Bernanke mentions in a footnote: "the MPC's focus on the forecast as a communication device raises the question of whether the forecast is an input to the policy decision or a joint product with the policy decision (meaning that the MPC sees the goal of the forecast to be as much rationalizing as informing the policy decision)". Thomas observes that "If the core of the process is something that is potentially both input and output,

[4] This is also referred to as an "optimal path" for interest rates, in the sense that it is one that the Committee itself believes is most consistent with minimizing its loss function.

Figure 28. Policy rate forecast

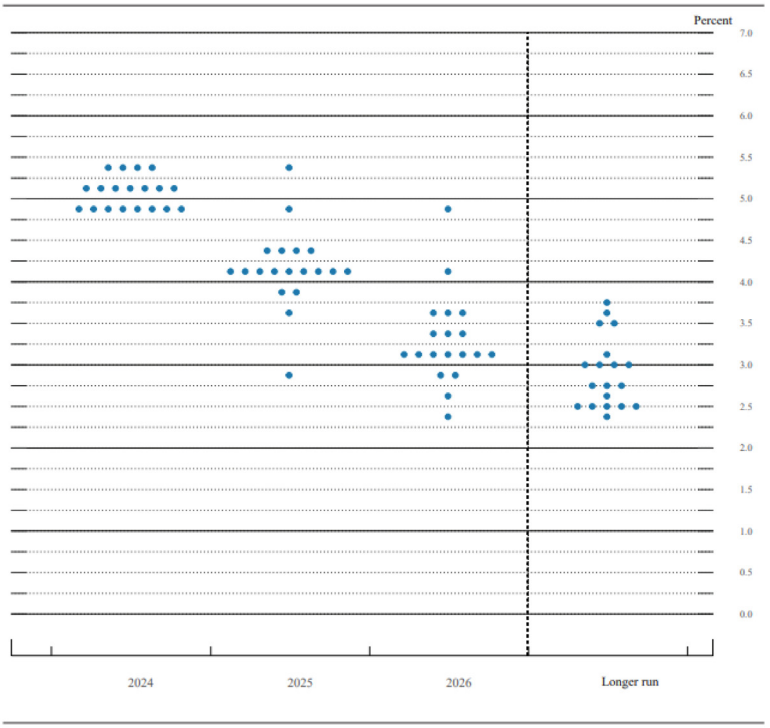
Per cent



Note. The solid line refers to outcomes, dashed lines refer to the Riksbank's forecasts. Outcomes are daily rates and forecasts are quarterly averages.

Source: Riksbank (2024).

Figure 2. FOMC participants' assessments of appropriate monetary policy: Midpoint of target range or target level for the federal funds rate



NOTE: Each shaded circle indicates the value (rounded to the nearest 1/8 percentage point) of an individual participant's judgment of the midpoint of the appropriate target range for the federal funds rate or the appropriate target level for the federal funds rate at the end of the specified calendar year or over the longer run.

Source: Federal Reserve Board (2024).

Fig. 2. Alternative presentations of the central bank's outlook for policy rates: (a) policy rate forecast from the Riksbank's June 2024 Monetary Policy Report; (b) The FOMC's "dot plot" from the June 2024 policy meeting.
Source: Federal Reserve Board (2024) and Sveriges Riksbank (2024)

Schrodinger style, then that must be ironed out before drawing any further conclusions".

Many Bank watchers see practical problems with the MPC owning the forecast and bemoan the lack of a recommendation in the Bernanke Review to shift to a staff-owned forecast. Jack Meaning argues that the current practice "creates a forecast that no single MPC member believes in". In contrast, a staff forecast would "allow MPC members to focus on why they disagree or deviate from the staff's view, using the forecast as a neutral, internally consistent base to launch discussion from". Similarly, Francesca Monti contends that such a move would "create a clearer distinction between model outputs and judgments, which would help in monitoring their respective performance". And Jumana Saleheen speaks of a need to "close the disconnect between the model and the published projections", arguing that "the staff should own their forecast, so that the process of judgment and modelling are brought closer".

Reflecting on her time on the Committee, DeAnne Julius offers the interesting critique that "over time, the repeat game of negotiating a 'best collective judgment' every quarter adds to the problems of incrementalism and 'group think'". She strongly advocates moving to a staff forecast on the grounds that it "would free up Committee members' time, make better use of their links to the outside world, provide more real-time input, promote a broader debate within the MPC meetings and thereby improve the decision outcomes, especially in times of major external shocks". Sushil Wadhwani is of a similar opinion: "[A staff forecast] would allow the MPC to spend more time on debating the issues of the day versus being bogged down by the detail relating to econometric modeling, especially if much committee time will now be expended on the new model. Moreover, if the MPC is no longer responsible for the forecast, then its membership could more easily be widened to include a more diverse set of experiences", he writes.

3.5. The Bank's modeling framework

The Bernanke Review is scathing in its assessment of the state of the Bank's forecast and modeling framework.[5] We learn that "key software is out of date and lacks important functionality" and that "insufficient resources have been devoted to ensuring that the software and models underlying the forecast are adequately maintained (updated, stress tested, and periodically re-estimated.). Many of the watchers commented on this unfortunate situation. David Hendry and John Muellbauer ask, "What was the Bank thinking (and doing) that such basic advice on data management and model development is needed?" Jens Larsen queries "how can a process that involves so many highly qualified, hard-working, and well-intentioned staff and policymakers prove so change resistant, despite clear short-comings, extensive debate and with peer central banks taking different approaches?" and Chris Giles asks "how on earth did the BoE's management and governance arrangements allow its modeling to get into such a mess?" Rob Wood is not alone in concluding that more regimented and detailed parliamentary scrutiny of the Bank will be required going forward "to ensure the central bank does not get into this position again".

The baseline economic model ("COMPASS") comes in for particular criticism.[6] This model has "significant short-comings", the review informs us, and "its role in constructing the forecast has diminished considerably". It is disconcerting to learn of an "inability to capture fully some key channels of monetary transmission", that COMPASS "is no longer used to predict the effects of changes in interest rates or asset prices on the economy", and that "the shape of the forecast is not significantly constrained by the a priori theoretical properties of this model". COMPASS appears to be primarily "a framework for aggregating the output of other models and human judgements and to ensure that key accounting relationships among variables are maintained". Unsurprisingly, the Bernanke Review advises that a "new central model (or at a minimum a thorough revamping of COMPASS) will likely be needed". Rob Wood highlights that, in light of these revelations, Bank watchers can no longer rely on the COMPASS model multipliers as an accurate description of how the MPC thinks and how it responds to data. He argues that "an update on what the Bank thinks about these key inputs into its policy decision" is therefore urgently required.

Bank watchers had a range of views on the approach this model development project should take. David Hendry and John Muellbauer contend that the key problem is with "the New Keynesian DSGE formulation on which COMPASS was also based", which in their view is "fundamentally flawed". They go on to note that the Bank of England is now in "a small minority of central banks without [a semi-structural macroeconomic model], a model type that has become widely adopted, especially since the financial crisis". David Aikman agrees, arguing that "Dynamic General Equilibrium models like COMPASS and its predecessor BEQM are inherently more difficult to maintain and develop than models with looser micro-foundations". He observes that "there's an important lesson to learn from the fact [the FRB/US model, i.e., a model with looser micro-foundations than COMPASS] has been in use for around 30 years – updated and refined rather than replaced". However, Richard Barwell points out that "there were reasons why the Bank switched from the MTMM [a semi-structural model] to BEQM", namely "to obtain a well-defined steady state, stock-flow and flow-flow consistency and model-consistent rational expectations for all variables (not just inflation) to name but three".

Other commentators see the potential for the Bernanke Review to yield a step change in the quality of the forecast data infrastructure. Richard Davies advocates automating the process of feeding data into the forecast: "Models that start with an API call ensure comparable data is being

[5] For a review of the processes involved in generating macroeconomic forecasts, see Stekler (2007).

[6] For a description the COMPASS model, see Burgess et al. (2013).

used, save time, remove errors, and add transparency. Over time, the Bank should move to ensure that all of its analysis (charts, tables) related to the forecast are run from live APIs. The Bank would be a natural home or lead partner in a publicly available UK macroeconomics API". Moreover, "a microdata API should be developed, which despite being more complex than the macro version, would be a huge win for the Bank, and UK in general, as it would draw in researchers from across the world". Ben Nelson sees a future in which "a more seamless flow of data, model development and model outputs would broaden and deepen analytical expertise across the staff, in turn enhancing its ability to inform and challenge the MPC, making for better policy decisions over time".

Several commentators admonish the Bank for its lack of transparency about its forecast model infrastructure. Adrian Pagan notes that "for some years I have been looking at the BoE web page to find what model is being currently used in the projection. The only reference I could find was to the COMPASS paper of 2013". Mick Grady suggests the Bank should publish "a detailed description of the main forecasting model, including the key modelling assumptions contained within it and the multipliers/ impulse responses to a variety of real, nominal and policy shocks. The model code and data should also be made available to allow for external analysis".

The Bank may have the very best of intentions in responding to the Bernanke review and improving the forecasting toolkit, but it may face real obstacles to making significant progress. Dr Bernanke observes that "staff members with extensive experience in a particular domain generally have deeper knowledge of their subject area and can engage policymakers with more confidence. At the Federal Reserve and other major central banks, it is not uncommon for forecast leaders and subject matter specialists to have decades of experience in their area". But those specialists are not easy to recruit and retain. David Aikman questions whether the Bank has "the human resource capacity needed to maintain and develop the new framework" given that "the Bank's remuneration package is not at all competitive; as a result, it finds it incredibly difficult to recruit and retain the best and brightest". Meanwhile, Chris Giles queries how the specialist staff that the Bank does have are currently being deployed: "the majority of PhD researchers in these areas were found to spend little or no time on the core function of the central bank. This raised the question how the governance of the BoE had not noticed that it is employing expensive economists who seem to be using the Bank as a vehicle for academic research without the hassle of teaching students?"

Finally, Jumana Saleheen reminds us that the Bank has a well of external expertise it can draw on in addressing these failings. "To stay on the frontier", Jumana writes, "the Bank and MPC should set up a forecast advisory board with academics and practitioners. The Federal Reserve has an academic advisory board. The ECB regularly engages with academics and practitioners. Such dialogue promotes debate and awareness of forecasting best practice". Sanjay Raja goes further by arguing that external forecasters could participate in MPC briefing sessions, ahead of the forecast round. These proposals to increase external engagement in the forecast process have the potential to enhance objectivity, increase diversity of thought, and counter "group think" influences, while also ensuring that the Bank remains up to date with the cutting edge of research and practice.

4. Conclusions

This paper summarizes reactions to the Bernanke Review from a large number of expert UK macroeconomists, including those working in academia, financial markets, and business. Based on our discussions with Bank watchers and a review of the academic literature, we identify three critical topics that warrant further attention from the IJF community.

First, there is an ongoing debate about the relative merits of Dynamic Stochastic General Equilibrium (DSGE) models versus alternative modeling frameworks for forecasting macroeconomic variables. While the literature on DSGE models is extensive, more work is needed to explore how alternative frameworks, such as agent-based models or hybrid approaches, might complement or improve upon existing methods, particularly in capturing non-linear dynamics and rare events. It seems especially important for the academic community to engage with this debate at this juncture as various central banks are in the process of developing so-called semi-structural models along the lines of the FRB/US model for forecasting processes.[7]

Second, we see a growing need for practical guidance on scenario design and implementation. While there is a substantial body of literature on the theoretical underpinnings of scenarios, there is less clarity on how central banks should generate, communicate, and use scenarios in practice, particularly in the context of public engagement. This is compounded by the tension between communicating uncertainty and meeting the public's desire for clear guidance. There is also a need to develop methodologies geared toward evaluating the veracity of chosen scenarios ex post.

Third, the profession would benefit from greater exploration of institutional questions that sit at the intersection of forecasting and governance. For example, how can central banks reconcile publishing a forecast based on the MPC's preferred rate path with current market prices? And who should own the central bank's macroeconomic forecast: policymakers or the staff? These questions are both theoretically and practically significant, yet they remain underexplored in the academic literature.

We hope this paper motivates the IJF community to explore the topics highlighted by the Bernanke Review and the perspectives of the Bank watchers community in greater depth, fostering a stronger connection between academic research and the practical challenges of macroeconomic forecasting.

[7] See Brayton, Laubach, and Reifschneider (2014) for a description of the FRB/US model.

CRediT authorship contribution statement

David Aikman: Writing – review & editing, Writing – original draft. **Richard Barwell:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Aikman, D., & Barwell, R. (2024). The bernanke review: Responses from bank of England watchers. Available on the King's College London website: https://www.kcl.ac.uk/business/assets/pdf/qcgbf-working-papers/the-bernanke-review-into-the-bank-of-england's-forecast-process-responses-from-bank-of-england-watchers.pdf.

Bernanke, B. (2024). Forecasting for monetary policy making and communication at the Bank of England: a review. Available from the Bank of England website: https://www.bankofengland.co.uk/independent-evaluation-office/forecasting-for-monetary-policy-making-and-communication-at-the-bank-of-england-a-review/forecasting-for-monetary-policy-making-and-communication-at-the-bank-of-england-a-review.

Brayton, F., Laubach, T., & Reifschneider, D. (2014). The FRB/US model: A tool for macroeconomic policy analysis. *FEDS Notes*, https://www.federalreserve.gov/econresdata/notes/feds-notes/2014/a-tool-for-macroeconomic-policy-analysis.html.

Britton, E., Fisher, P., & Whitley, J. (1998). The inflation report projections: understanding the fan chart. *Bank of England Quarterly Bulletin Q1*, https://www.bankofengland.co.uk/quarterly-bulletin/1998/q1/the-inflation-report-projections-understanding-the-fan-chart.

Burgess, S., Fernandez-Corugedo, E., Groth, C., Harrison, R., Monti, F., Theodoridis, K., et al. (2013). The Bank of England's forecasting platform: COMPASS, MAPS, EASE, and the suite of models. *Bank of England Working Paper No. 471*, https://www.bankofengland.co.uk/-/media/boe/files/working-paper/2013/the-boes-forecasting-platform-compass-maps-ease-and-the-suite-of-models.pdf.

Bank of England (2002). *Inflation report*. https://www.bankofengland.co.uk/-/media/boe/files/inflation-report/2002/may-2002.pdf.

Bank of England (2019). *Monetary policy report*. https://www.bankofengland.co.uk/-/media/boe/files/monetary-policy-report/2019/november/monetary-policy-report-november-2019.pdf.

Federal Reserve Board (2024). *Monetary policy report*. https://www.federalreserve.gov/monetarypolicy/files/20240705_mprfullreport.pdf.

Goodhart, C. (2009). The interest rate conditioning assumption. *International Journal of Central Banking*, *5*(2), 85–108.

Hendry, D., & Pretis, F. (2023). Analysing differences between scenarios. *International Journal of Forecasting*, *39*(2), 754–771.

Stekler, H. O. (2007). The future of macroeconomic forecasting: Understanding the forecasting process. *International Journal of Forecasting*, *23*(2), 237–248, http://www.sciencedirect.com/science/article/B6V92-4N4S0CC-3/1/eb8a502d2b79e3add4e668bde0beb69c.

Svensson, L. (1997). Inflation forecast targeting: implementing and monitoring inflation targets. *European Economic Review*, *41*(6), 1, 111–46.

Sveriges Riksbank (2024). *Monetary policy report, 2024*. https://www.riksbank.se/globalassets/media/rapporter/ppr/penningpolitiska-rapporter-och-uppdateringar/engelska/2024/monetary-policy-report-june-2024.pdf.